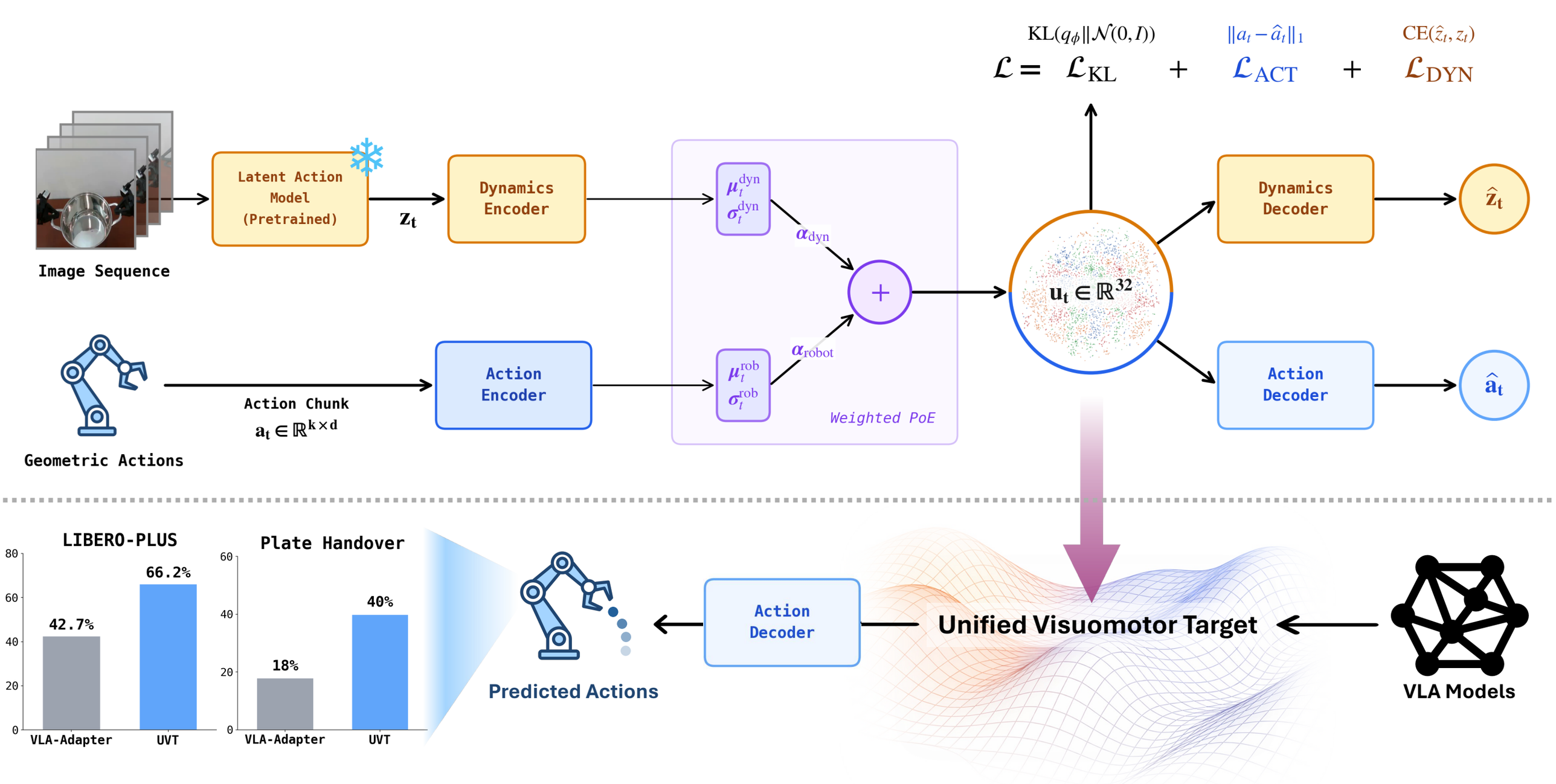
$\mathcal{L} = \mathcal{L}_{KL} + \mathcal{L}_{ACT} + \mathcal{L}_{DYN}$
$KL(q_\phi \| \mathcal{N}(0,I))$
$\|a_t - \hat{a}_t\|_1$
$CE(\hat{z}_t, z_t)$
Image Sequence
Latent Action Model (Pretrained)
z_t
Dynamics Encoder
μ_t^{dyn} σ_t^{dyn}
α_{dyn}
α_{robot}
μ_t^{rob} σ_t^{rob}
Weighted PoE
Geometric Actions
Action Chunk
$a_t \in \mathbb{R}^{k \times d}$
Action Encoder
$u_t \in \mathbb{R}^{32}$
Dynamics Decoder
$\hat{z}_t$
Action Decoder
$\hat{a}_t$
LIBERO-PLUS
42.7%
66.2%
VLA-Adapter
UVT
Plate Handover
18%
40%
VLA-Adapter
UVT
Predicted Actions
Action Decoder
Unified Visuomotor Target
VLA Models